

GAME TEST REPORT

SlotForge One

Pre-Certification Self-Assessment

PASSED WITH NOTES

Report Number	BS-SLOTS-20260801-98ADDBD7
Date of Report	2026-08-01
Issued By	BlueStuff Game Studio · Automated Test Engine
Report Recipient	6a43beed9eae7b49adcdc60e
Engine Under Test	slots
Standard Tested Against	GLI-19 v3.0 — Interactive Gaming Systems
Classification	Pre-certification self-assessment (Non-Jurisdictional)
Software Fingerprint	sha256:b7861063fdb9bc4fb98a0a18c7253564f1fd967b... · 34 files
Result Summary	3 passed · 0 failed · 1 warnings · 0 manual · 0 source findings

Disclaimer. This is an automated pre-certification self-assessment produced by BlueStuff Game Studio's own test engine, published unmodified — including any failures. It is tested *against* the GLI-19 v3.0 technical standard but is **not** an accredited-laboratory certification and does not imply affiliation with, or endorsement by, Gaming Laboratories International. Final certification requires an independent accredited testing laboratory.

1. Executive Summary

Every hard requirement that ran passed. Some results are provisional or need human review.

Evaluator's Assessment

The reasoned opinion of the automated evaluator, based strictly on the measured results in this report.

GLI-19 Compliance Evaluation — Game 7121cfc6 (Slots / slotrocket-5c5a)

Run: 98addbd7 · Seed gli-1785584726535 · 2026-08-01

Overall Determination: Pass-with-Reservations

The RNG subsystem is clean across all three applicable clauses. The §4.7.1 RTP result for the primary engine is provisional rather than conclusive, and a second registered engine variant produced zero payouts — meaning full certification cannot be issued until both issues are resolved.

RNG & Statistical Integrity (§3.2.2, §3.2.3, §3.2.4) — PASS

All seven GLI-19 §3.2.2 tests were applied across 600,000 draws and 5 independent replicates. No test was rejected at the 99% family-wise confidence level (Holm–Bonferroni; Fisher combined $p = 0.762$). Distribution uniformity (§3.2.3) is confirmed for both float and integer outputs. Sequential independence (§3.2.4) holds across serial, overlaps, interplay, and per-round outcome tests (Fisher combined $p = 0.975$). The crypto-backed shared RNG is functioning correctly and these results are reproducible from the stated seed.

Return-to-Player (§4.7.1) — WARN — Blocks Certification

Two issues were found:

- **example engine — zero payouts over 200,000 spins.** This variant returned no player payouts across the entire sample. It is either a stub, a misconfigured registration, or a broken payout path. GLI-19 §4.7.1 requires a minimum 75% theoretical RTP; a zero-payout engine trivially fails that threshold. It cannot be certified as-is.
- **slotforge-mvp — RTP 91.178% ±6.261pp (provisional).** The measured RTP is above the 75% floor, which is encouraging, but the ±6.26 percentage-point confidence interval is wide enough to be inconclusive. At 200,000 spins this result does not establish the theoretical RTP to the precision GLI certification requires. The run must be widened before the value can be reported as confirmed.

For real players, an under-sampled RTP means the operator cannot guarantee the advertised return, and a zero-payout engine variant — if reachable in production — represents a complete failure of player protection.

Adversarial Findings — None

No adversarial probes produced findings in this run. Note that runtime probes (concurrent double-spend, cash-out-after-settlement, session fixation) require a containerized backend; if those were not executed against a live instance, those attack classes remain untested and should be flagged as out-of-scope for this run rather than assumed clean.

Recommendations (Priority Order)

1. **[Blocking] Remove or fully implement the `example` engine.** If it is a placeholder, deregister it before submission. If it is a real game variant, its payout logic must be implemented and audited separately.
2. **[Blocking] Widen the `slotforge-mvp` RTP run.** Increase spins to at least 2,000,000 (preferably 10,000,000) to narrow the confidence interval to $\pm 1\text{--}2\text{pp}$ and establish a certifiable theoretical RTP. Re-run with the same seed family for reproducibility.
3. **[Recommended] Execute runtime adversarial probes** against a containerized backend instance (double-spend, post-settlement cash-out, session fixation) to complete the §4.6 fairness and integrity picture.
4. **[Informational] RNG results are strong** — no further action needed on §3.2.2–3.2.4. These can be carried forward without re-running if the RNG codebase is unchanged.

2. Scope & Evaluation Elements

The following elements were evaluated to the extent applicable to automated testing of this software:

Element	Status
Software & System Version Control	Performed
Source Code Review	Performed
Game Accounting (RTP / payout)	Performed
RNG & Statistical Analysis (GLI-19 §3.2)	Performed
Functionality Testing	Partial

3. Requirement Results

Requirement	Result	Sample	Statistic
3.2.2	PASS	600,000	$p = 1.69\text{e-}1$
3.2.3	PASS	600,000	$p = 4.68\text{e-}1$
3.2.4	PASS	320,000	$p = 5.21\text{e-}1$
4.7.1	WARN	400,000	—

4. Requirement Detail

3.2.2 PASS

Shared RNG source (crypto-backed, used by every engine on this template). float() scaled to 32 bins, 5 independent draws combined by Fisher's method. 7 of the 7 tests named in §3.2.2 applied. None rejected at the 99% family-wise confidence level (Holm–Bonferroni; Fisher combined $p = 7.622e-1$).

3.2.3 PASS

RNG output is uniformly distributed (float 32-bin: Fisher chi-square = 9.70, $p = 0.4676$; int 64-range: $p = 0.9365$).

3.2.4 PASS

Independence of the shared RNG (serial, overlaps, interplay) and of each engine's per-round outcomes (slotforge-mvp). 4 of the 7 tests named in §3.2.2 applied. None rejected at the 99% family-wise confidence level (Holm–Bonferroni; Fisher combined $p = 9.747e-1$).

4.7.1 WARN

Return-to-player measured per registered engine by driving play() with the template RNG: • example: no payouts observed over 200,000 spins — stub or non-payout engine; nothing to certify. • slotforge-mvp: RTP 91.178% $\pm 6.261pp$ over 200,000 spins (hit 35.03%, max 2775.0x) — provisional (widen sample). GLI-19 §4.7.1 requires a theoretical payout of at least 75%. (1 non-payout/stub engine(s) noted above.)

5. Source Code Review

No adversarial findings. The probes that were run did not surface an exploitable issue.

6. Software & System Version Control

A SHA-256 checksum was generated for each of the 34 source files in the engine under test. The aggregate fingerprint binds this report to exactly the code tested.

Aggregate fingerprint: sha256:b7861063fdb9bc4fb98a0a18c7253564f1fd967bcd25db790cdfd7d17235caf6

File	Bytes	SHA-256
mock-rgs/server.js	6,941	efd3b571471ff8715098...
package-lock.json	39,470	80396a8b9dc627a8b7bb...
package.json	650	dbe06c33c23778839840...
src/adapters/InlineWallet.js	3,591	0f3d448ea065d1965b13...
src/adapters/RgsAdapter.js	11,741	bf249f586850cb72fe3d...
src/adapters/WalletAdapter.js	1,827	96abb0fa90468bbcb94a...
src/app.js	2,960	4853c5de79c0d57c1eb9...
src/config/index.js	2,048	70ca00b00159577b15f3...
src/engines/GameEngine.js	3,261	83d6fff9314cb1845d45...
src/engines/example/ExampleEngine.js	2,827	b146f8322209c3764e3c...
src/engines/index.js	1,548	0b61a5e98c81add4284c...
src/engines/slotforge-mvp/SlotForgeEngine.js	19,265	d1e0ebee4f7d106838d1...
src/engines/slotforge-mvp/analytics.js	7,427	b709c62a10f07eee7b5b...
src/engines/slotforge-mvp/defaultMath.js	17,834	6873b7255c0b10641b26...
src/middleware/errorHandler.js	936	e3c1912b7fe62a86a9cd...
src/models/index.js	4,175	763a60ca94e6cec820e5...
src/routes/config.js	957	a4c5adc3cd5cceaabe52...
src/routes/game.js	1,742	07682a3da73b014775f1...
src/routes/session.js	1,309	f3d8ea81cc657ab65445...
src/server.js	1,010	cda1ea92956411d4dd94...
src/services/AuditService.js	1,205	2887a68af4ba2500ecb7...
src/services/GameService.js	13,281	2111fc9a04afb7c07507...
src/services/SessionManager.js	3,501	9a133289e6702199ceb7...
src/store/index.js	554	90e0279b64368867bc05...
src/store/memoryStore.js	1,524	33fcc249eed7130402f9...
src/store/mongoStore.js	1,177	89d67140ee3156976ce5...
src/utils/errors.js	1,710	0d4ee388580860da1aba...

src/utils/ids.js	1,036	19816c83e246299f080e...
src/utils/logger.js	1,938	6ab4fc4d1af74e62a7a4...
src/utils/money.js	996	6b633695f0ace232e6cd...
src/utils/rng.js	1,106	6a75832219a1d9540812...
tests/lifecycle.test.js	11,972	e4c304aebdb95a95b6f1...
tests/rgs-adapter.test.js	8,944	d75b50b824c88f8ab9e7...
tests/slotforge-mvp.test.js	44,552	e352794f8295f43f63be...

7. Verification & Reproduction

The test engine has no dependencies and no LLM in its numeric path — every statistic is computed by reviewable in-tree code. This run reproduces with:

```
node src/cli.ts --engine=slots --repo=D:\Projects\Devops\MCP\mcp-server\.repo-cache\games\bluestuff-
io__smokestack-original-games\BackendEngines\slotrocket-5c5a --samples=100000 --spins=200000 --
replicates=5 --seed=gli-1785584726535
```

The seven statistical tests are the ones GLI-19 §3.2.2 names explicitly (chi-square, overlaps, coupon collector, runs, interplay correlation, serial correlation, duplicates), evaluated collectively at 99% family-wise confidence and combined across independent replicate seeds.

8. Test Environment

Test engine	SportsITDev/gli-audit-engine
Runtime	Node v24.14.0
Seed	gli-1785584726535
Started	2026-08-01T11:45:26.689Z
Finished	2026-08-01T11:45:32.060Z
Generated by	gli-audit-engine game-tester agent